

Reviewed Preprint

v1 • December 5, 2025

Not revised

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Competing interests: Additional

Declarations: There is no competing interest.

Reviewing editor: Benjamin M

Hogan, Peter MacCallum Cancer Centre, Australia

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Junctional and Actomyosin Dynamics Drive Endothelial Cell Rearrangements during Vascular Tube Formation

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eLife Assessment

This **valuable** study focuses on a unique morphogenetic module, the junction-based lamellipodia (JBL). It provides a biomechanical understanding of how JBLs control endothelial cell-cell junctional remodelling to generate lumenised, multicellular blood vessels. The manuscript represents a robust, thoughtfully executed, and **convincing** study that uses high-resolution time-lapse imaging combined with pharmacological treatments to advance our understanding of lumen formation in vascular development.

<https://doi.org/10.7554/eLife.109264.1.sa3>

Abstract

Lumen formation is a key process during the morphogenesis of tubular organs such as the vertebrate vascular network. At the cellular level, lumen formation can be achieved by cell shape changes or cell rearrangements. We have previously shown that such cell rearrangements are driven by oscillating membrane protrusions, called junction-based lamellipodia (JBL), which provide a ratchet mechanism driving convergent cell movements to join local lumens and generate vascular patency. By performing *in vivo* time-lapse imaging at high spatiotemporal resolution, we have analyzed the cytoskeletal and junctional dynamics, which underlie JBL formation and function. We show that JBL formation requires the activity of the F-actin nucleation complex Arp2/3. We further show that a novel junction is formed at the distal end of the JBL from a pool of VE-cadherin, originating from outside the initial JBL area.

Subsequently, proximal and distal junctions merge and fuse, a process driven by actomyosin contractility. Prior to this fusion we observe a specific recruitment Myl9 within the interjunctional space. Furthermore, inhibition of actomyosin contractility abrogates junctional merging. Taken together, our analyses demonstrate that JBL constitute a module, which by alternate generation of pushing forces (JBL formation) and pulling forces (junctional merging) provide the physical means of endothelial cells to elongate and to rearrange thereby generating a continuous vascular lumen.

Introduction

Vascular networks are formed by an array of morphogenetic processes, such as sprout outgrowth, anastomosis and pruning, which depend on the tight coordination of dynamic endothelial cell behaviors (Betz et al., 2016 ; Ellertsdóttir et al., 2010 ; Phng, 2018 ; Schuermann et al., 2014 ). In order to transport blood, the newly formed blood vessels need to form a continuous lumen. Studies of lumen formation in mammalian and zebrafish embryos have revealed that the early blood vessels of the trunk are formed by different morphogenetic processes. The larger axial vessels, the dorsal aorta (DA) and posterior cardinal vein (PCV), form their lumens between coalescing endothelial cells, a process referred to as cord hollowing (Ellertsdóttir et al., 2010 ; Jin et al., 2005 ; Strilić et al., 2009 ; Zeeb et al., 2010 ). *In vivo* timelapse analyses of the formation

of the intersomitic vessels (ISV) and the dorsal longitudinal anastomotic vessels (DLAV) in zebrafish have revealed the existence of two alternative mechanisms of lumenization. In one mechanism, called transcellular lumen formation, the luminal cell membrane invaginates into the cell. This cell hollowing process is driven by blood pressure and results in so-called seamless tubes, which contain a lumen surrounded by single cells (Betz et al., 2016 [↗](#); Gebala et al., 2016 [↗](#); Herwig et al., 2011 [↗](#); Lenard et al., 2013 [↗](#); Phng and Belting, 2021 [↗](#)). Alternatively, local lumens, which are formed between individual cells within a sprout can be fused by cell rearrangements (Herwig et al., 2011 [↗](#); Sauter et al., 2014 [↗](#)). Endothelial cell rearrangements generate multicellular tubes (Betz et al., 2016 [↗](#); Phng and Belting, 2021 [↗](#)). Studies examining lumenization of ISV and DLAV have shown that transcellular lumen formation requires blood pressure and that, in the absence of blood flow, endothelial cell rearrangements appear to be the default mechanism of vascular tube formation (Herwig et al., 2011 [↗](#); Lenard et al., 2013 [↗](#)).

Endothelial cells are very motile - they can move within the angiogenic sprout and also in patent vessels in the presence of blood flow (Blum et al., 2008 [↗](#); Franco et al., 2016 [↗](#); Jakobsson et al., 2010 [↗](#)). Here, a dynamic balance between inter-endothelial cell adhesion and plasticity allows angiogenic sprouting while maintaining the endothelial seal. Previous analysis on blood vessel formation and anastomosis in zebrafish has shown that junctional remodeling is central to many aspects of morphogenetic endothelial cell-cell interactions. In particular, the adhesion molecule VE-cadherin (Cdh5) is essential for coordinated cell shape changes during multicellular tube formation, as loss of VE-cadherin was shown to inhibit cell rearrangements (Sauter et al., 2014 [↗](#)). Furthermore, rather than supporting a passive role of VE-cadherin, which is consistent with the maintenance of vascular integrity, an active force generating function for VE-cadherin in this process was suggested (Sauter et al., 2014 [↗](#)). By performing in vivo high-resolution time-lapse imaging and genetic analyses, we have shown that endothelial cells use oriented, oscillating junction-based lamellipodia (JBL) to move over each other and thus provide the physical means for cell rearrangements to drive multicellular tube formation (Paatero et al., 2018 [↗](#)).

To better understand the cellular and molecular mechanisms of JBL function we have undertaken a detailed analysis of junctional and actomyosin dynamics. Taken together, our data show that JBL provide forward movement by generating pushing and pulling forces at the poles of junctional rings thereby elongating the cell and promoting cell rearrangements.

Results

Junction-based lamellipodia generate luminal connectivity

Lumenization of vascular sprouts is central to blood vessel formation. Although different morphogenetic mechanisms of vascular tube formation have been described, in vivo analysis in zebrafish embryos has established cord hollowing as the default mechanism of blood vessel formation in the absence of blood flow (Herwig et al., 2011 [↗](#); Lenard et al., 2013 [↗](#)). During the cord hollowing process neighboring cells form local lumens which become interconnected by cell rearrangements (Andrew and Ewald, 2010 [↗](#)). In order to visualize the fusion of such luminal pockets we followed the localization GFP-Podxl, which labels the apical cell membrane, together with F-actin reporter UCHD-mRuby2, to delineate endothelial cell-cell junctions and F-actin base protrusions (Fig. 1 [↗](#)). Time-lapse recordings revealed that fusion of apical membrane compartments occurs within a few hours upon initiation of DLAV formation and is associated with junctional rearrangements between the three associated cells (Fig. 1b, c [↗](#) and Supplementary movie 2 [↗](#)). In agreement with our previous work, higher temporal resolution imaging showed that cell junctions formed transient F-actin-based protrusion (JBL) at the pole with respect to junctional ring elongation (Fig. 1c, d [↗](#) and Supplementary movie 1 [↗](#), and Supplementary movie 3 [↗](#)) (Paatero et al., 2018 [↗](#)).

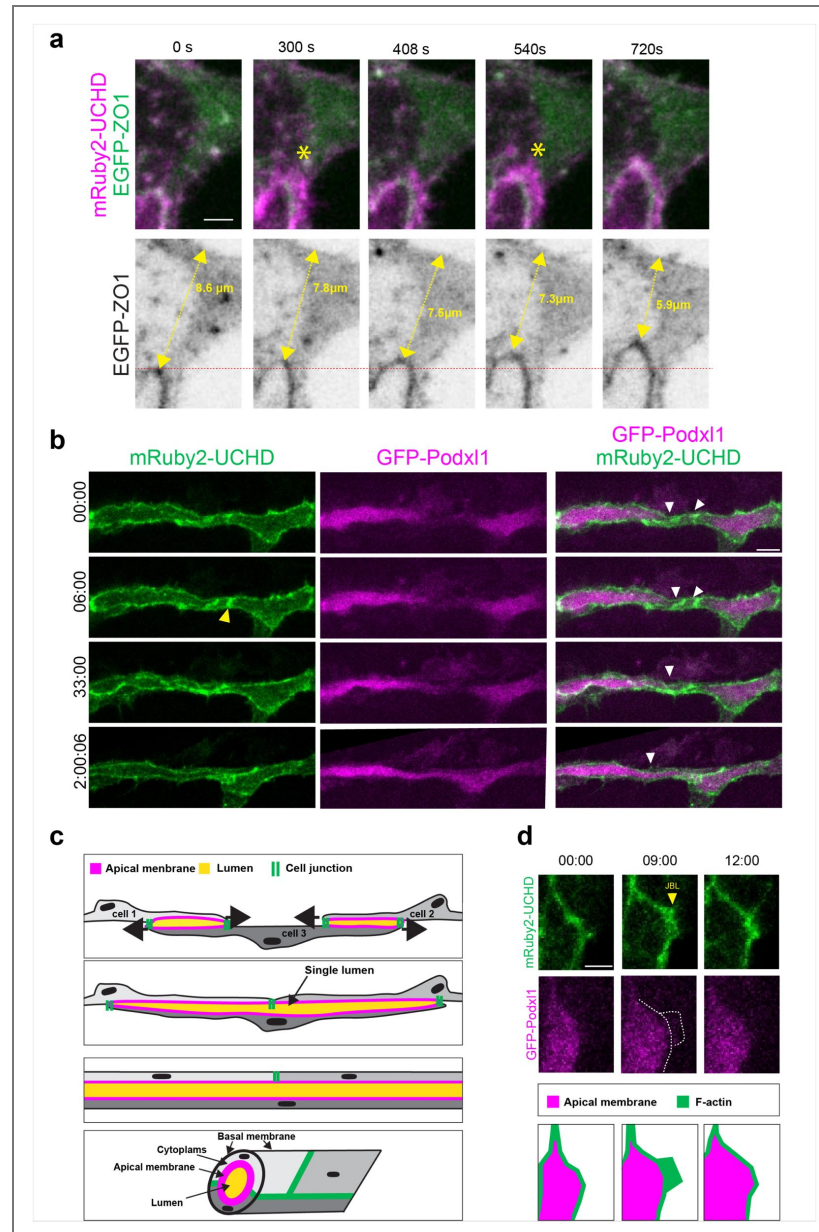


Figure 1. JBL generate luminal connectivity.

a: High resolution time lapse of EGFP-ZO1 and mRuby2-UCHD during junctional rearrangements. Yellow asterisks point JBL. yellow double-headed arrows indicate the distance between the junction and a reference point within the cell cytoplasm (Supplementary movie 1 [\[1\]](#)). **b:** Time-lapse stills of EGFP-Podxl2 and mRuby2-UCHD during luminal fusion, starting at approximately 32 minutes. The leading edges of the converging junctions are indicated by white arrowhead. Yellow arrowheads point to JBL. Scale bar 10 μ m (Supplementary movie 2 [\[2\]](#)). **c:** Graphic depiction of blood vessel lumenization by cell convergence. Top: At first, cells 1 and 3 and cell 2 and 3, respectively, are forming cell-cell interfaces (cell junctions: green), which enclose local lumens (apical membrane: purple). Middle: JBL drive convergent cell movements, which lead to the formation of a novel cell contact (cells 1 and 2) and the merging of the two lumens into one (bottom). Graphic depiction of a lumenized DLAV, displaying the multicellular architecture. **d:** timelapse movie (Supplementary movie 3 [\[3\]](#)) showing the oscillatory behavior of JBL (yellow arrowhead) during DLAV formation (30hpf). The white dashed line surrounds the UCHD-labeled JBL domain. Scale bar 5 μ m. Schematic reproduction of the time lapse in the bottom panel.

JBL form distal junctions by de novo recruitment of VE-cadherin

We have previously shown that JBL are oscillating protrusions, which drive junction elongation and thereby cell rearrangements during tube formation. A salient feature of JBL is the formation of a distal junction in front of the JBL (Paatero et al., 2018 [↗](#)).

To gain more insight into the dynamics of distal junction formation, we compared the distribution of F-actin (Ruby-UCHD) with that of VE-cadherin (VE-cad-Venus) and ZO-1 (EGFP-ZO1), respectively (Fig. 2 [↗](#)). Time-lapse experiments revealed distinct and dynamic localization of these proteins. During early phases of JBL formation VE-cadherin was diffusely distributed throughout the protrusion and accumulated in aggregates leading to a characteristic “double junction” pattern at the distal tip of the junctional ring (Fig. 2a [↗](#) and Supplementary movie 4 [↗](#)). Compared to VE-cadherin and ZO-1 the early demarcation of F-actin at proximal and distal junction was less sharp (Fig. 2b, c [↗](#) and Supplementary movies 5 [↗](#), 6 [↗](#)). However, early aggregates of VE-cadherin and ZO-1 were always positive for Ruby-UCHD, showing that these three proteins were present from the onset of distal junction formation. At subsequent stages, as the distal becomes more mature, ZO-1 and VE-cadherin at the proximal junction were maintained, while the Ruby-UCHD signal started to fade (Fig. 2c-e [↗](#)).

Formation of the distal junction may occur in different ways. On one hand, it may form by a local junctional disassembly, followed by a reaggregation at the distal end of the JBL. This would essentially consist of a junctional rearrangement process. On the other hand, the distal junction may form de novo by recruitment of VE-cadherin from pools outside of the proximal junction. In order to distinguish between these scenarios, we performed color conversion experiments (Supplementary Fig. 1 [↗](#) and Supplementary Movie 7 [↗](#)). Pools of photoconvertible VE-cadherin (Cdh5-mClav2) were differentially labeled by exposure to UV light within a narrow region of interest at the pole of the junctional ring (Supplementary Fig. 1a [↗](#)), which resulted in efficient green-to-red conversion (Supplementary Fig. 1b [↗](#)). When we followed subsequent JBL dynamics over time, we noted that converted VE-cad was absent from the protrusion as well as the distal junction (Supplementary Fig. 1c [↗](#) and Supplementary movie 7 [↗](#), 5 experiments). Instead, the distal junction contained non-converted VE-cadherin, indicating that the distal junction was formed *de novo* by recruitment of VE-cadherin from pools outside the proximal junction.

Arp2/3 complex dynamics spatiotemporally correlates with JBL formation

Junctional protrusions are commonly seen in developing and newly formed blood vessels in the zebrafish embryo. Cell culture experiments, using human umbilical cord endothelial cells (HUVEC), have shown that such lamellipodial structures can control monolayer integrity and vascular permeability (Alves et al., 2018 [↗](#); Martinelli et al., 2013 [↗](#)). In particular, so-called junction-associated intermittent lamellipodia (JAIL) are prominent in sub-confluent HUVEC, where they serve to repair junctional gaps (Taha et al., 2014 [↗](#)). In contrast to JAIL JBL are highly polarized relative to the vessel axis and they drive cell rearrangements rather than maintaining vascular integrity (Paatero et al., 2018 [↗](#)). However, both lamellipodial structures are similar in their emergence from and association with endothelial cell junctions.

JAIL formation appears to be triggered by a local reduction in junctional tension and requires the activation of the ARP2/3 complex. We used a transgenic zebrafish line expressing Arpc1b-Venus in the vasculature to test whether Arp2/3 also regulates JBL formation. Time-lapse analysis showed a discrete spatiotemporal deployment of Arp2/3 at the leading edge of the forming JBL (Fig. 3 [↗](#)). Imaging JBL over several oscillations revealed that Arp2/3 accumulation coincided with the initiation and the elongation of JBL, while it disappeared at the end of each JBL cycle (Fig. 3a [↗](#) and Supplementary movie 8 [↗](#)). Comparison with ZO-1 showed that Arp2/3 localized distally with no overlap within the proximal junction (Fig. 3a [↗](#)). When we imaged Arp2/3 together with mRuby2-UCHD at higher temporal resolution (30s/frame), we observed that during JBL extension

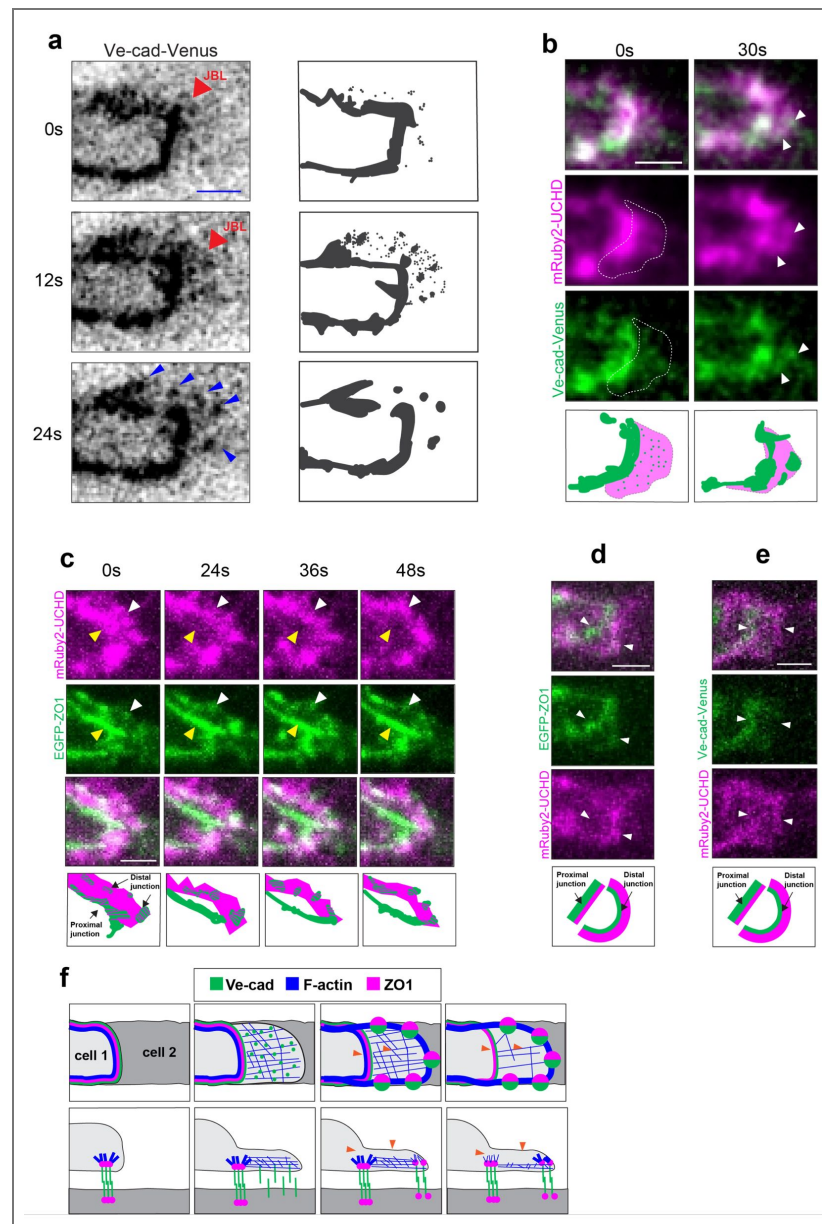


Figure 2. JBL form new junctions at the distal end of the actin protrusion.

a: Time-lapse of VE-cad-Venus imaged at rate of 1stack/12s during distal junction formation. (supplementary movie 4 [\[link\]](#)) New distal junctions emerge in clusters at the distal end of the JBL. Ve-cad is diffusely localized during early step of JBL, while it accumulates in big foci at later time points. Similar observations were made in 9 separate movies. Scale bar 2 μ m. On the right panels we present a simplified reproduction of labeled VE-cad spatiotemporal distribution during distal junction formation. **b:** Time-lapse of mRuby2-UCHD and VE-cad-Venus during JBL (0s) and distal junction formation (30s). (supplementary movie 5 [\[link\]](#)). The dashed line encircles the protrusion. White arrow heads point distal junction foci at the distal tip of the protrusion. Scale bar 2 μ m. Similar observations were made in 15 movies. **c:** Time-lapse of EGFP-ZO1 and mRuby2-UCHD during the presence of proximodistal junction. After distal junction formation, f-actin gradually diminishes from proximal junction and interjunctional space, while maintaining strong localization at the distal junction. Similar observations were made in 5 movies. White arrowheads confine UCHD expression domain. Scale bar 2 μ m. **d,e** Still image of EGFP-ZO1 and mRuby2-UCHD (D), and mRuby2-UCHD and VE-cad-Venus (E) respectively, in proximodistal junction. (supplementary video 6 [\[link\]](#)) While junctional proteins localize more strongly at the proximal junction, respect to the new, immature distal junction, f-actin faintly localize at the proximal junction and strongly at the distal. Similar observations were made in 10+11 videos. Scale bar 2 μ m. **f** Schematic representation the spatiotemporal distribution of Cdh5, F-actin and ZO-1 during formation of the distal junction in top and side view: Initially Cdh5 appears diffusely dispersed throughout the membrane protrusion. Distal junction foci form at the distal side of the lamellipodia. Actin network along the distal junction. Finally, junctional actin network along the proximal junction dissolves, as pointed by orange arrow heads.

Arp2/3 was maintained at the leading edge (Fig. 3b, c [↗](#) and Supplementary movie 9 [↗](#)). Taken together, these data show that Arp2/3 specifically marks the distal edge of the JBL during formation and elongation (Fig. 3d [↗](#)).

As the above findings suggest that Arp2/3 is involved in JBL formation, we interfered with Arp2/3 function by pharmacological inhibition using CK666 (Hetrick et al., 2013 [↗](#)). We used acute treatments to avoid secondary effects and performed live-imaging on rearranging endothelial cell junctions. Application of 0.2 mM CK666 at 32 hpf completely disrupted junctional Arp2/3 localization within 30 min. (Supplementary Fig. 2 [↗](#)). Analysis of F-actin organization revealed an increase of filopodia emanating from the junction in the area, which usually was occupied by JBL (Fig. 4a-c [↗](#) and Supplementary movies 10 [↗](#), 11 [↗](#)). The abundance of these filopodia prevented the localization of JBL at the junctional poles consistent with previous cell culture studies (Henson et al., 2015 [↗](#)). This suggests that filopodia are formed instead of lamellipodia in the absence of Arp2/3 activity. To test consequences of the loss of Arp2/3 activity on cell rearrangements, we measured the speed of junctional elongation in control and CK666 treated embryos. By imaging ZO-1 labeled junctions over 1 hour we found that the speed of junction elongation was strongly reduced from an average of 8 $\mu\text{m}/\text{h}$ to 3 $\mu\text{m}/\text{h}$ (Fig. 3d,e [↗](#) and Supplementary movies 12 [↗](#), 13 [↗](#)). Taken together the results show that JBL formation depends on Arp2/3 function and that the absence of Arp2/3 activity leads to a strong impairment of endothelial cell rearrangements during blood vessel formation.

Proximal and distal junctions resolve by junctional merging and fusion

The appearance of “double junctions” (consisting of a proximal and a distal junction) is a distinctive junctional feature observed in JBL. Their spatiotemporal dynamics suggest that they represent part of the mechanism by which JBL promote junction elongation. Live-imaging revealed a cyclic turnover between the two junctional states: single junction and double junction pattern. Time-lapse imaging of VE-cadherin-Venus at high temporal resolution showed that this turnover is mediated by the cyclic formation of the distal junction followed by the merging and fusion of the proximal and distal junctions (Fig. 5a [↗](#) and Supplementary movie 14 [↗](#)).

Myosin light-chain localization correlates with junctional merging

The cyclic junctional pattern in JBL suggests that it may represent a physical ratchet mechanism, which drives junction elongation. Therefore, we wanted to explore whether junctional mergence could be driven by actomyosin contractility. To monitor the spatiotemporal distribution of actomyosin contractility during JBL cycle, we utilized myosin regulatory light chain (MLC) (Myl9a-GFP and Myl9b-Cherry) as reporters for non-muscular myosin II (NMII) activity (Lancino et al., 2018 [↗](#)). Myl9 accumulated along the junctions but was strongly enriched in JBL at both ends of the junctional ring (yellow arrowheads in Fig. 5b [↗](#)).

During cell rearrangements JBL are formed at each pole of the junctional ring in a complementary fashion: each JBL is formed by one of the interacting cells corresponding to the forward orientation with respect to cell movement (Paatero et al., 2018 [↗](#)) (see schematic in Fig. 5d [↗](#)). To test whether Myl9 accumulation corresponds to the forward orientation of the JBL, we analyzed rare mosaic situations in which Myl9a-GFP was expressed in only one of the partnering cells. Indeed, we observed high levels of Myl9a-GFP in JBL formed by the high-expressing cell pointing into the forward direction of the elongating junction (Fig. 5d, e [↗](#)).

Vascular tube formation by cell rearrangements has also been described in other vascular beds. We turned to the posterior cerebral vein (PCeV), which is generated by JBL-driven cell rearrangements converting the initial tube architecture from unicellular to multicellular (Lenard et al., 2013 [↗](#)). When we compared Myl9a-EGFP distribution to mRuby2-UCHD, we found that the former localized within the JBL adjacent to the forming distal junction of the JBL (see yellow arrowheads in Fig. 5f [↗](#) and Supplementary movie 15 [↗](#)).

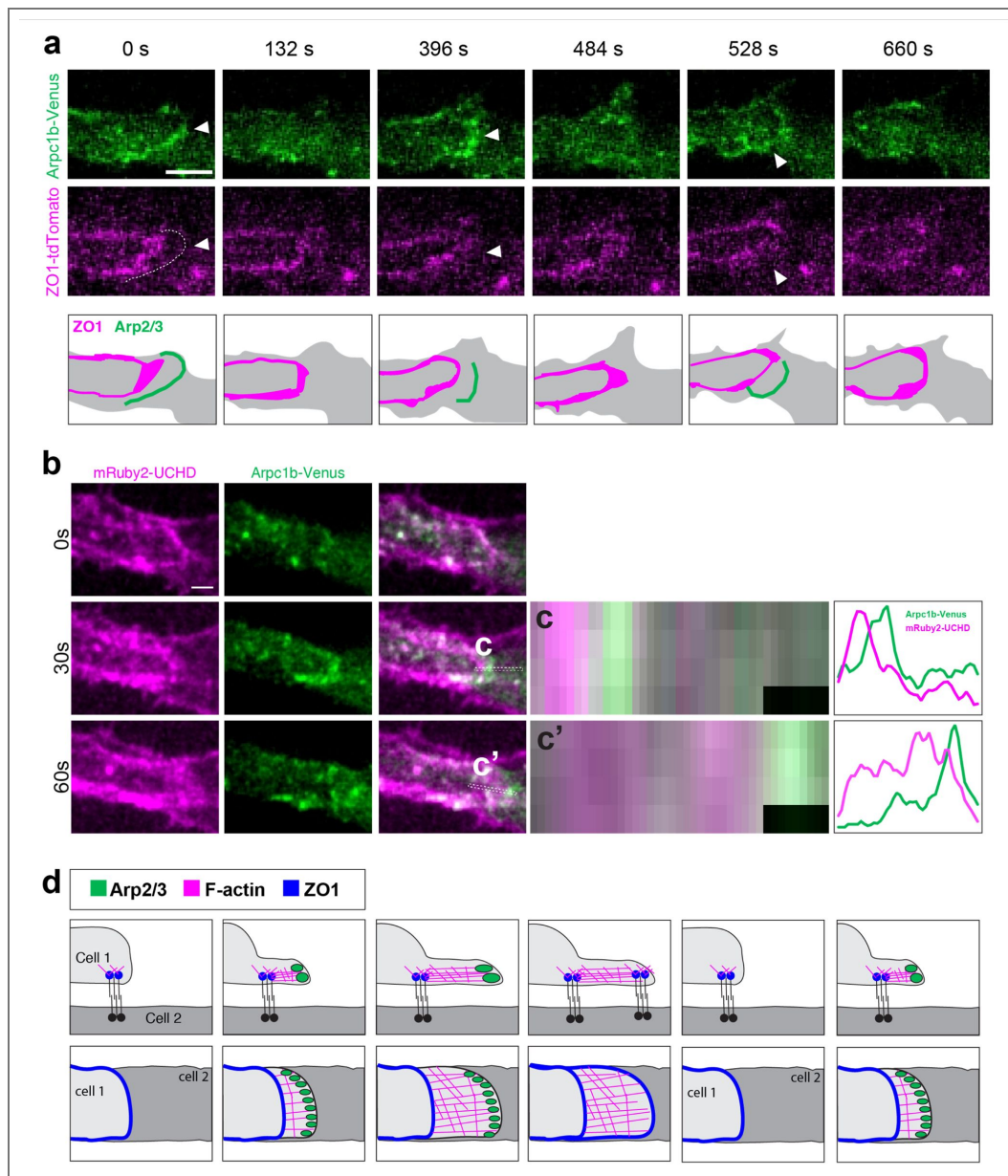


Figure 3. Arp2/3 localization oscillates at the distal end of JBL.

a Time-lapse (Supplementary movie 8) of Arpc1b-Venus and ZO1-tdTomato at about 30 hpf, showing deposition of Arp2/3 at the distal side of the junctional ring during 2 JBL cycles. Scale bars 2 μ m. **b** Time-lapse (Supplementary movie 9) of Arpc1b-Venus and mRuby2-UCHD during a JBL event. Scale bar 2 μ m. **c** and **c'** Plots showing the average intensity of Arpc1b-Venus and mRuby2-UCHD along rectangular ROIs drawn along the JBL at 30 and 60 seconds, respectively. **d** Schematic model of top and side view of spatiotemporal localization of ARP2/3, f-actin and ZO1 during JBL formation and extension for 1.5 JBL cycle.

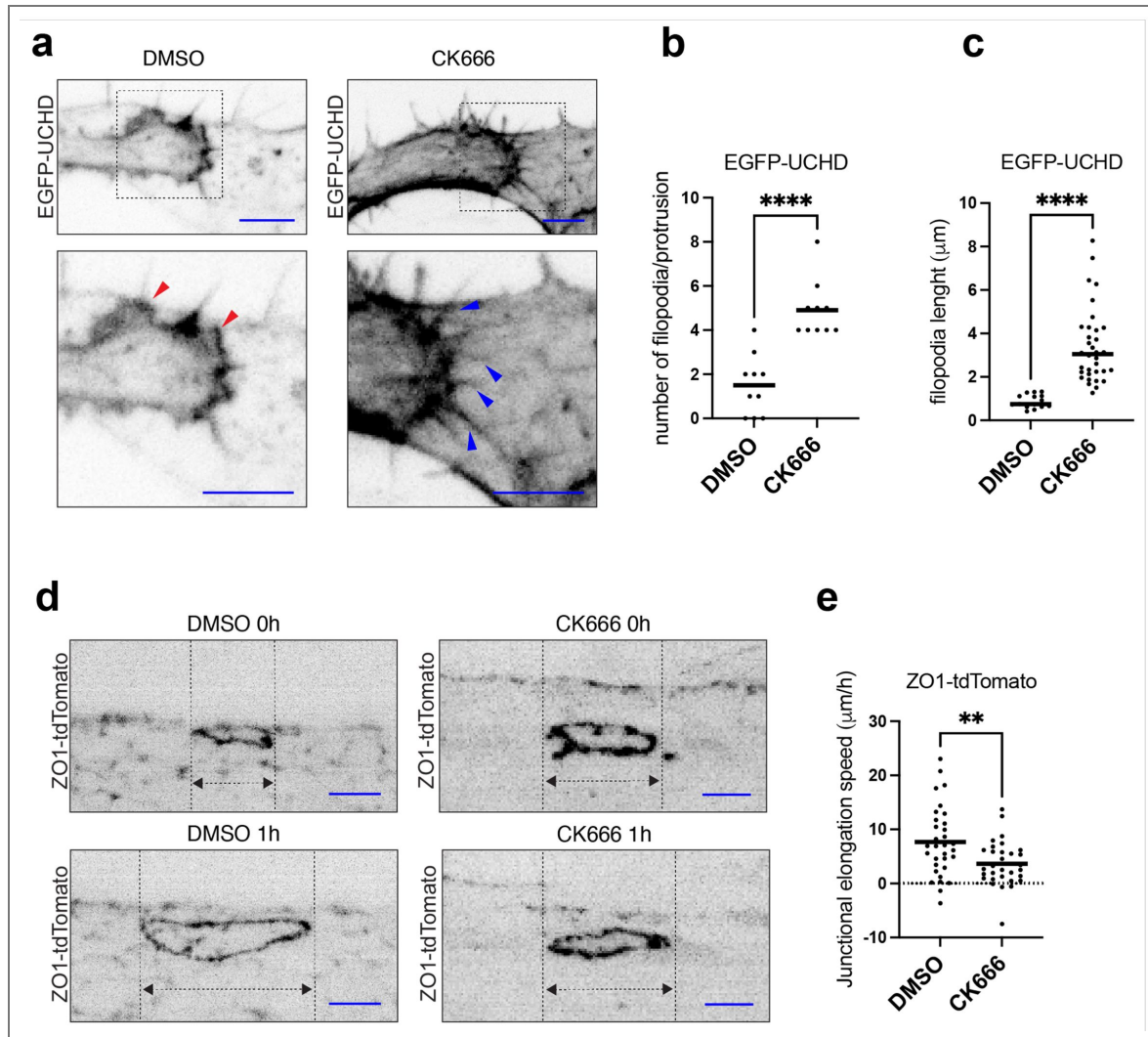


Figure 4. Arp2/3 is required for junctional elongation and JBL formation.

a Still-images from a movie (Supplementary movies 10 [DOI](#), 11 [DOI](#)) of JBL labelled with EGFP-UCHD, at around 32 hpf, in the presence of DMSO (1%) or 1h incubation with CK666 (200 μM). Similar observations were made in 10 movies. Scale bar 5 μm . Similar observations were made in more than 12 movies. Scale bar 5 μm . **b** Quantification of number of filopodia/JBL in DMSO (1%), $n=10$ JBL and CK666 (200 μM), $n=10$ JBL events. Unpaired t-test was used for statistical analysis. (P value < 0.0001). **c** Quantification JBL filopodia length in DMSO (1%), $n=15$ JBL and CK666 (200 μM), $n=34$ JBL events. **d** Still images from time-lapse movies (Supplementary movies 12 [DOI](#), 13 [DOI](#)) of an ZO1-tdTomato labeled junctional rings in the presence of DMSO (1%), CK666 (200 μM). Top panels $t=0$ and bottom panels after 1 h incubation. Scalebar 5 Scale bar 10 μm . **e** Quantification of the junctional elongation velocity in DMSO (1%), $n=21$ junctions (10 embryos) and CK666 (200 μM), $n=24$ (10 embryos). Dotted line indicated no movement observed, black lines are medians. Unpaired t-test was used for statistical analysis. (P value = 0.0047).

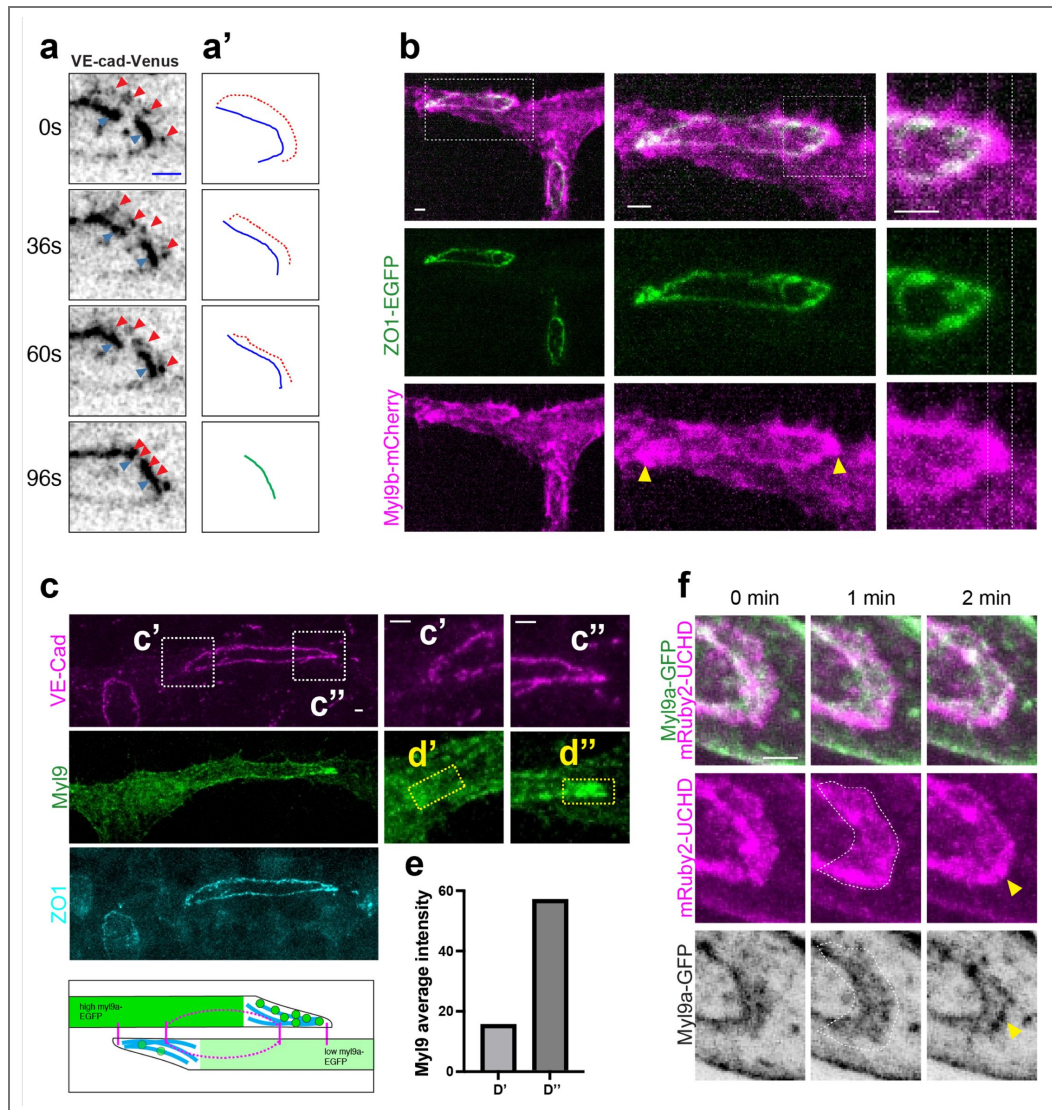


Figure 5. MLC is enriched at the junction pole.

a: Time lapse (Supplementary movie 14) of VE-cad-Venus during junctional merging. Blue and red arrowheads point proximal and distal junction respectively. The 2 junctions are gradually moving closer until they merge. Similar observations were made in 13 movies. Scale bars 2 μ m. **b:** Confocal stack of ISV and DLAV of Myl9b-mCherry and ZO1-EGFP expressing embryo at around 32 hpf. Yellow arrow heads point JBLs. Dashed lines underline Myl9b-mCherry distal expression domain at the junction pole. Scale bars 2 μ m. **c:** Confocal stack of DLAV and ISV immunostaining against GFP (green), VE-cad (magenta) and ZO1 (cyan) of an embryo expressing myl9a-GFP at around 32hpf in a case of mosaic expression: the left cell has a strong Myl9-GFP expression (as in the schematic). **c'** and **c''** dashed squares delimitate JBL regions of the 2 cells. **e** quantification of Myl9-EGFP average intensity the yellow dashed squares ROIs **d'** and **d''** drawn on the JBL domain. The average intensity inside the ROIs was corrected subtracting the cytoplasmic background. The cytoplasmic average background was calculated drawing 5 squares in non-junctional cytoplasmic region of the highly Myl9-GFP expressing cell. **f.** Still images of a Myl9a-GFP and mRuby2-UCHD during JB event anastomosis in PCeV at around 60 hpf Imaged with spinning disc (Supplementary movie 15). White dashed lines delimitate JBL region. Yellow arrowheads point distal junction. Myl9-GFP is enriched inside the lamellipodia and localize at the distal junction in later time points. Scale bars 2 μ m.

In order to gain a better understanding how actomyosin contractility may drive junctional merge, we imaged the distribution of Myl9 at high spatiotemporal resolution. We first used VE-cadherin-EGFP to label proximal and distal junctions as spatial landmarks and thus demarcate the area of interest for the quantification of Myl9b-mCherry in space and time (Fig. 6a, b, Supplementary Fig. 3, and Supplementary movies 16, 17, 18, 19, 20). We defined three areas – proximal junction, interjunctional and distal junction and quantified Myl9b-mCherry at two time-points (0" and 45") at the onset and during junctional merge, respectively (Fig. 6b). These quantifications reveal a redistribution Myl9 from the proximal and distal junctions towards the interjunctional space (Fig. 6c). This increase indicates a strong recruitment of Myl9 and a concomitant increase of actomyosin contractility within the area between the proximal and distal junctions.

Actomyosin contractility is required for junction elongation and junctional merge

In order to assess the role of actomyosin contractility in junction elongation more directly, we used the Rock inhibitor Y-27632 (Uehata et al., 1997) and measured the length of junctional rings over 60 minutes. Rock inhibition strongly inhibited junction elongation compared to DMSO controls (Fig. 7a, b and Supplementary movies 21, 22). To test whether the lack of junction elongation under Rock inhibition may be caused by defects in JBL function, we performed acute short-term Y-27632 treatment and performed live-imaging over 30 minutes to image JBL dynamics during reduction of actomyosin contractility – with a focus on the merging of proximal and distal junctions. To this end, we imaged 20 (Y-27632) and 13 (DMSO) events of junctional merging, respectively, and tracked individual interjunctional distances over time (Fig. 7c-e and Supplementary movies 23, 24). Inhibition of actomyosin contractility led to a strong inhibition of junctional merge, ranging from a reduction in speed to a complete block in junctional merge. This blockage in junctional merge was reflected by the overall increased occurrence of “double junctions” under Rock inhibition (20 (Y-27632) vs 13 events (DMSO) in 18 videos). Taken together, these results show that high levels of actomyosin contractility reside within interjunctional space and that these contractile forces are responsible for junctional merge. Furthermore, these data suggest that JBL formation together with junctional merge are the driving force of junctional elongation and endothelial cell rearrangements during angiogenic blood vessel formation.

Discussion

Lumen formation is key in the formation of tubular organs such as blood vessels. Endothelial cells can use different morphogenetic pathways to generate a lumen, including membrane invagination, cord hollowing or lumen ensheathment (Eberlein et al., 2021; Phng and Belting, 2021). The dorsal longitudinal anastomotic vessel (DLAV) of the zebrafish embryo can be formed by either membrane invagination or cell rearrangements, which result in unicellular or multicellular tubes, respectively (Betz et al., 2016; Phng and Belting, 2021). Here, we have analyzed the cell rearrangements, which connect small luminal pockets within the initial endothelial rod, thereby generating a continuous lumen in a fully patent vascular tube.

The coalescence of local, extracellular lumens is commonly observed during the cord hollowing process of different biological tubes, including the notochord of *Ciona intestinalis*, the vertebrate dorsal aorta as well as the neural tube and intestine (Bagnat et al., 2007; Dong et al., 2009; Gladysheva et al., 2021; Jin et al., 2005; Strlič et al., 2009). In those tubular organs, in which lumen coalescence was analyzed, it was found that it is mainly caused by lumen expansion, which in turn is driven by directed liquid transport and hydrostatic pressure (Bagnat et al., 2007; Lowery and Sive, 2005; Munson et al., 2008; Navis and Bagnat, 2015). In contrast, the lumen coalescence observed in the zebrafish ISV, DLAV and PCeV is accomplished by active cell intercalation. Here, endothelial cells actively elongate their cell-cell junctions until they form novel cell-cell contacts, which allow the connection of the two adjacent lumens.

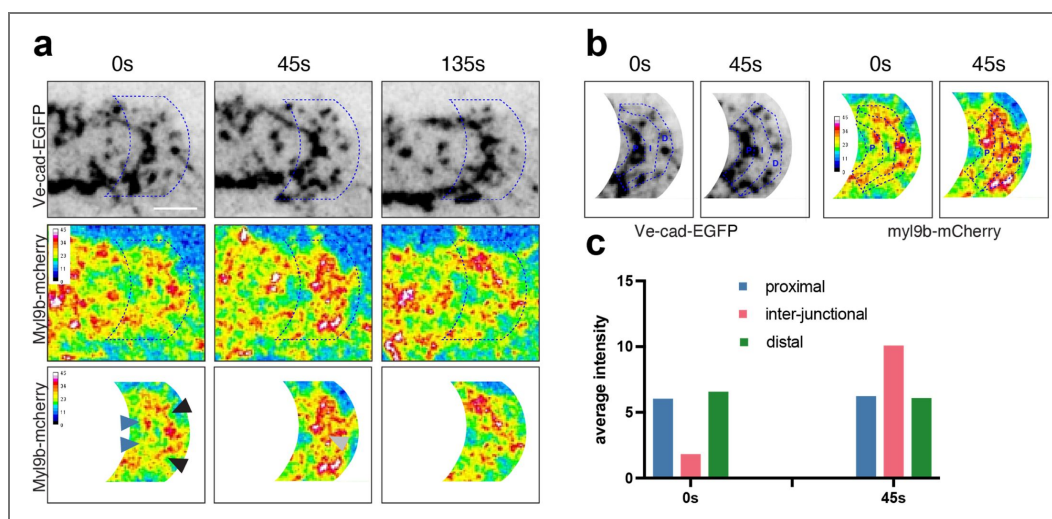


Figure 6. Myosin light-chain dynamics correlates with junctional merging.

a Time-lapse (Supplementary movie 16) of Cd5ΔC-EGFP and Myl9b-mCherry during junctional merging. Blue dash lines confine the applied mask. Blue and black arrowheads point to MLC accumulation on proximal and distal junction. Similar observations were done in 6 videos. Grays arrowheads point interjunctional newly recruited Myl9-mCherry. Similar observations were made in 9 videos. Scale bars 2 μm. **b** Dashed blue lines delimitate 3 regions: proximal junction region (P), distal junction region (D) and interjunctional space (I). **c** Plot of average intensity of Myl9b-mCherry in the 3 regions, over time. The intensities have been corrected for background which has been evaluated as average of average intensities in 5 rectangles in the cell cytoplasm.

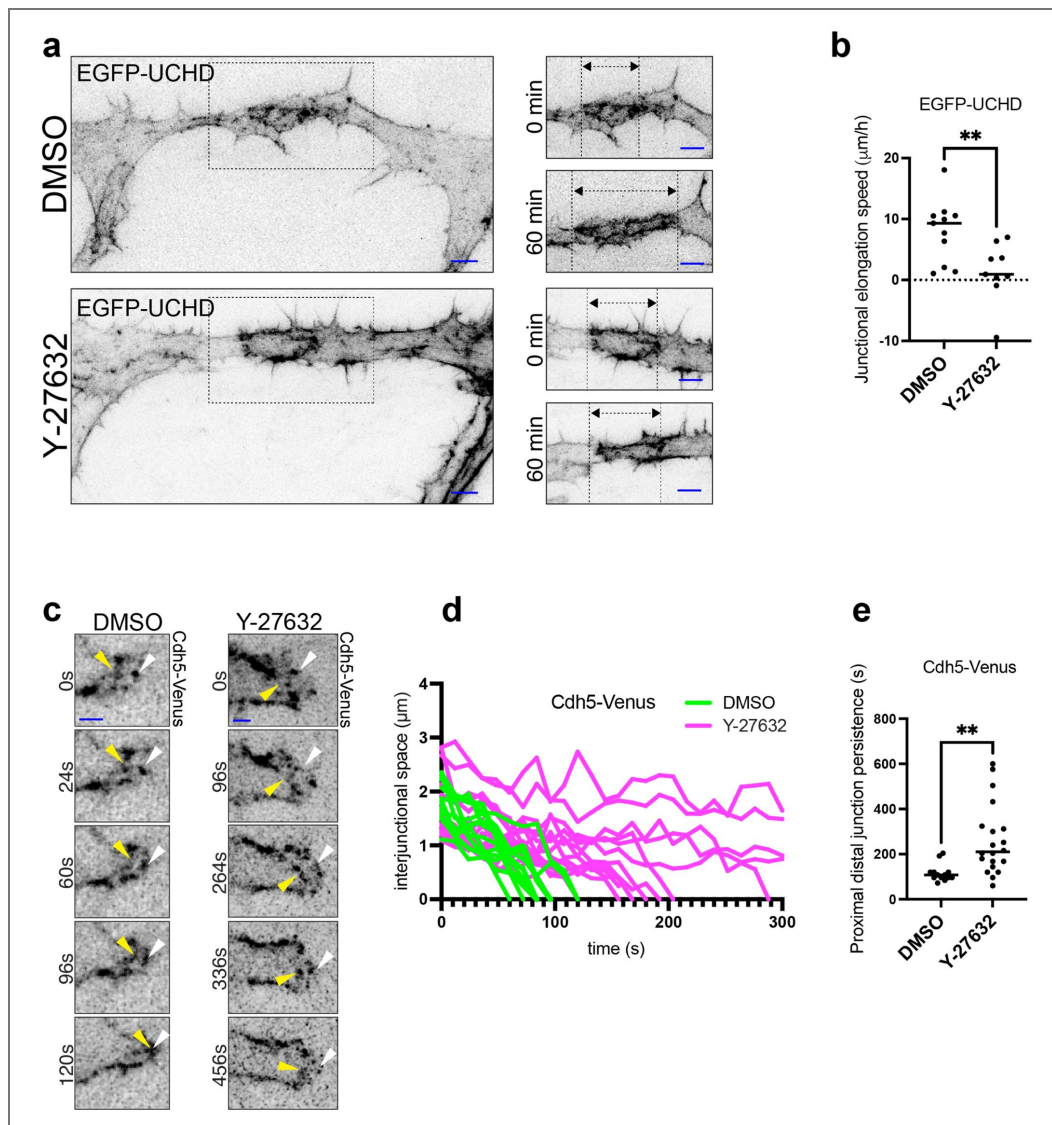


Figure 7. Actomyosin contractility drives junctional conversion.

a Still images from movies (Supplementary movies 21 and 22) EGFP-UCHD labelled junctional rings around 32 hpf, in the presence of DMSO (1%) or Y27632 (45 μM). Top panels $t = 0$ and bottom panels 60 min incubation. **b** Quantification of the junctional elongation velocity in DMSO (1%), $n = 11$ junctions and CK666 (200 μM), $n = 8$. Dotted line indicated no movement observed, black lines are medians. Unpaired t-test was used for statistical analysis. P value = 0.0081. **c** Time-lapses (supplementary movies 23 and 24) VE-cad-Venus labelled proximal-distal junction, during junctional merging in the presence of DMSO (1%) or Y27632 (75 μM). White and yellow arrowheads are pointing distal and proximal junctions respectively. **d** Tracking of proximal-distal junction distance over time of single junctional merging events in DMSO (1%) (green lines), and Y27632 (75 μM) treated embryos (magenta lines). **e** Quantification of the persistence of proximal and distal junction in DMSO (1%), $n = 13$ junctional merging events and Y27632 (75 μM) $n = 20$. Unpaired t-test was used for statistical analysis. P value = 0.0027.

We have previously shown that endothelial junction elongation is driven by junction-based lamellipodia (JBL) (Paatero et al., 2018). Careful analysis of junctional and cytoskeletal reporters revealed that JBL dynamics can be subdivided into 4 discrete steps: 1) JBL formation, 2) formation of a new, distal junction, 3) junction conversion and 4) junction stabilization (Paatero et al., 2018). Taken together, these studies showed that JBL provide an oscillating ratchet-like mechanism, in which endothelial cells employ F-actin-based protrusions and VE-cadherin-mediated cell-cell contacts to move over each other. JBL thus provide the physical means for cell rearrangements to drive multicellular tube formation.

In this study we have further explored the junctional and actomyosin dynamics to discern how JBL generate the motive force required for junction elongation and cell rearrangements. In essence, our data show that JBL require ARP2/3 dependent F-actin polymerization for JBL formation as well as actomyosin contractility to merge the proximal and distal junctions at the end of the JBL cycle. Thus, by sequential deployment of pushing (F-actin polymerization) and pulling (junction merging) forces, JBL promote forward movement of endothelial cells (summarized in Fig. 8).

Oscillatory junctional protrusions of endothelial cells have also been described in cultured HUVECs (Cao et al., 2017; Taha et al., 2014). Here, so-called junction-associated intermittent lamellipodia (JAIL) form in similar intervals as JBL. However, differences in junctional dynamics suggest that JAIL and JBL represent related but different cellular activities. Notably and in contrast with JAIL, JBL form a distal junction at the leading edge of the protrusion. Our studies show that these “double junctions” are an integral part of the cellular mechanism of junction elongation. We have previously shown that JBL are most prominent during multicellular tube formation and become less prevalent and smaller thereafter (Paatero et al., 2018). It is therefore likely that JAIL and JBL have different functions during vascular development. While JBL are required for polarized, directed junctional remodeling, JAIL are involved in junctional maintenance and endothelial integrity. Lamellipodial structures resembling JBL, displaying double junctions, have recently been described in the oak leaf-shaped endothelial cells of dermal lymphatics in the mouse (Schoofs et al., 2025). Here, the double junctions are thought to regulate endothelial shape and in particular to stabilize the interendothelial cell overlap, which is important to control the interstitial fluid transport.

Our photoconversion experiments (see Supplementary Fig. 1) show that the distal junction is formed *de novo* from VE-cadherin pools originating from outside the existing (proximal) junction. During distal junction formation, we observe that VE-cadherin forms clusters resembling spot-junctions observed in the context of organ morphogenesis in other systems. Consistent with our observation such spotjunctions have been shown to be important in the formation of contractile F-actin fibers in these systems (Lecuit et al., 2011). While we are currently not able to monitor actin flow in our experimental *in vivo* system, our observation that the formation of the distal junction coincides with the formation of a distal F-actin belt is consistent with a clutch mechanism, in which translocation of VE-cadherin triggers reorganization of the cortical actin cytoskeleton similar to what has recently been described in cell culture studies (Noordstra et al., 2023). Furthermore, our previous findings from zebrafish *cdh5* null mutants or embryos expressing a VE-cadherin protein (VE-cad Δ CT) lacking the β -catenin binding site show that VE-cadherin mediated cell-cell interactions as well as VE-cad/F-actin interaction are important for JBL-driven junction elongation (Paatero et al., 2018; Sauter et al., 2014).

The cell motility promoting activity of VE-cadherin during tube formation may be quite unique, considering that in most studies cadherins have been described to stabilize epithelia, limit cell motility and to regulate contact inhibition of locomotion (Campàs et al., 2024; Maître and Heisenberg, 2013; Noordstra et al., 2023). Cadherin-based cell interactions have been shown to be essential for dynamic cell movements in several morphogenetic processes (Friedl and Mayor, 2017; Mayor and Etienne-Manneville, 2016). In the case of collective border cell migration during *Drosophila* oogenesis and migration of primordial germ cells in zebrafish, cadherins have been shown to provide traction for collective and single cell migration, respectively (Blaser et al., 2006; Cai et al., 2014).

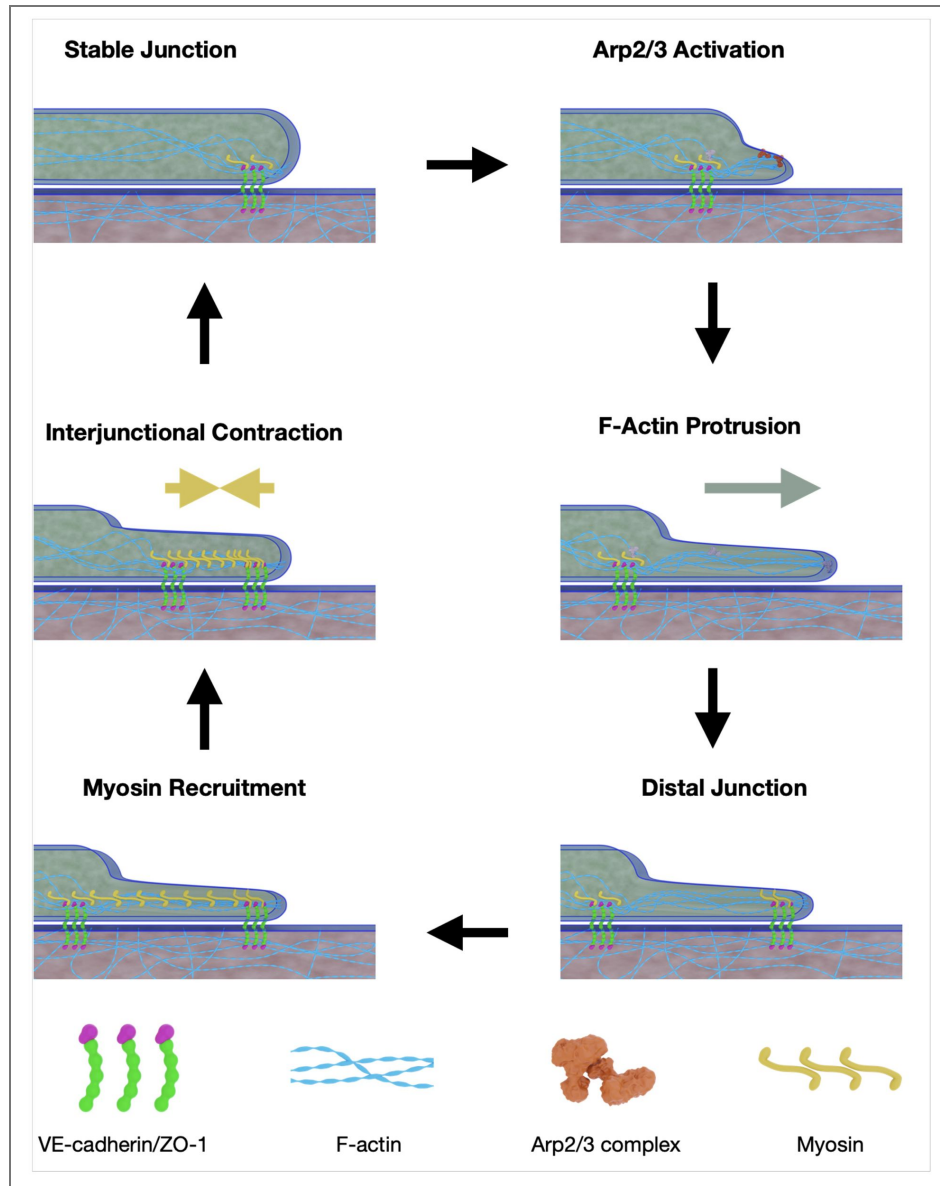


Figure 8. Schematic representation of different steps during JBL oscillation.

(see [supplementary Movie 25](#)) JBL formation is initiated by Arp2/3 activation. The JBL is pushed forward by F-actin polymerization. At the distal end a new cell-cell junction is formed. MyosinII is recruited to the interjunctional space. Actomyosin contraction merges the proximal and distal junctions resulting in an overall elongation of the junctional ring.

A salient feature of JBL is their appearance at both ends of each rearranging cell. This bipolar activity causes the cells to elongate in both directions and rearrange rather than to migrate. Furthermore, and in contrast to lamellipodia-driven mechanisms of cell migration, the domain of actomyosin contractility is confined within the interjunctional area (between distal and proximal junctions, see Figs. 6 [↗](#) and 8 [↗](#)). This suggests that the restricted area of contractility is not only demarcated by the proximal and distal junctions, but that these junctions act as insulators to segregate compartments of different actomyosin tension along the cellular interface. We propose that this insulator function is essential to permit junction elongation.

In this study, we have investigated the cellular and molecular basis of force generation in JBL. Through detailed analysis of junctional, cytoskeletal, and regulatory proteins, we uncover a biphasic mechanism involving alternating protrusive and contractile forces. Notably, and in contrast to classical paradigms of cell migration, we show that actomyosin contractility is spatially restricted to a narrow zone at the junctional pole during endothelial cell movement. Together, our findings reveal a novel mechanism of cell rearrangement during vascular cord hollowing. JBL serve as localized modules that generate both protrusive and contractile forces during junctional remodeling and cell elongation. Moreover, JBL-localized junctions may act as insulators, spatially confining actomyosin tension to the junctional pole.

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Low melting point agarose	Sigma-Aldrich	CAT# A9414

Lead contact

Further information and request for resources and reagents should be directed and will be fulfilled by the Lead Contact; Heinz-Georg Belting (heinz-georg.belting@unibas.ch).

Materials availability

Materials and zebrafish lines generated and used in this study are available upon request to the lead contact.

Fish maintenance and stocks:

Zebrafish (*Danio rerio*) were maintained according to FELASA guidelines (Aleström et al., 2020). All experiments were performed in accordance with federal guidelines and were approved by the Kantonales Veterinäramt of Kanton Basel-Stadt (1027H, 1014HE2, 1014 G).

The following zebrafish lines were used in this study:

Name	Origin
<i>Tg(fli:Gal4ff)^{ubs3}</i>	(Herwig et al., 2011)
<i>Tg(UAS:mRuby2-UCHD)^{ubs20}</i>	(Paatero et al., 2018)
<i>Tg(UAS:EGFP-UCHD)^{ubs18}</i>	(Sauteur et al., 2014)
<i>TgBAC(ve-cad:ve-cadTS)^{uq11bh}</i>	(Lagendijk et al., 2017)
<i>Tg(UAS:cdh5ΔC-EGFP)^{ubs60}</i>	this study, (Lenard et al., 2013)
<i>TgKl(tp1a-tdTomato)^{pd1224}</i>	(Levic et al., 2021)
<i>Tg(fli1a:Arpc1b-mVenus-p2A-Arpc3-mTurq2)^{mr25}</i>	(Malchow et al., 2024)
<i>Tg(kdrl:Myl9a-GFP)^{ip5Tg}</i>	(Lancino et al., 2018)
<i>TgBAC(cdh5:cdh5-mClavGR2)^{ubs58}</i>	(Yin et al., 2024)

Sex was not considered in this study because zebrafish embryos used for the experiments were at developmental stages prior to sex differentiation. Therefore, the sex of the embryos was not determined or relevant to the outcomes.

Live imaging of zebrafish embryos

In DLAV experiments, zebrafish embryos between 30 to 32 hpf were dechorionated and anaesthetized with 0.16 mg/ml (1×) tricaine methanesulfonate (Sigma). The embryos were selected for presence of fluorescence and mounted into microwell dishes within 0.55% low-melting-point agarose (ROTI) and covered with E3 buffer containing 1× tricaine. In PceV experiments, transgenic embryos selected for presence of fluorescence were anaesthetized in 1× tricaine (0.08%) and mounted in a 35 mm glassbottom Petri dish (0.17 mm, MatTek), using 0.7% low melting agarose

(Sigma) containing 0.08% tricaine and 0.003% 1-phenyl-2-thiourea (PTU; Sigma-Aldrich). The embryos were mounted into microwell dishes within 0.55% low-melting-point agarose (ROTI) and covered with E3 buffer containing 1× tricaine.

A Leica SP5 confocal microscope with a 40× water immersion objective was used for live-imaging experiments. High spatiotemporal resolution time-lapse images were acquired with an Olympus SpinSR spinning disc microscope using a 60X silicon oil (NA = 1.3) objective, with a z-stack step size of 0.6 μm at different frequencies depending on the experiment.

Photoconversion experiments

Photoconversion experiments were performed using a Leica SP5 confocal microscope with a 40× water immersion objective (NA = 1.1). Photoconversions were applied to manually selected regions of interest (ROIs) with a 405 nm laser for 10–20 seconds to induce conversion, monitored by simultaneous imaging at 488 nm to ensure complete conversion. All the imaging was performed at 28.5°C.

Junction elongation assay

Junctional elongation was assessed by measuring elongation speed of isolated junctional rings during DLAV formation between 30 and 32 hpf. Inhibitor treatments CK666 (200 μM), Y-27632 (75 μM), or DMSO (1%) were applied 1h before mounting of embryos into 0.55% low-melting point agarose. The same concentrations of chemicals were applied to the low-melting-point agarose mounting medium and the E3 medium on top of it before imaging and imaging the junctions for 60–90 min on a Leica SP5 (Y-27632) Olympus SpinSR spinning disc microscope or Olympus SpinSR spinning disc (CK666) microscope.

Junctional merging tracking

Speed of junctional merge was evaluated by monitoring isolated junctional rings during DLAV formation. Inhibitor treatment Y-27632 (75 μM) or DMSO (1%) were applied 30 min before mounting. The same concentrations of chemicals were applied to the low-melting-point agarose mounting medium and the E3 medium on top of it before imaging and imaging the junctions for 10–15 min on Olympus SpinSR spinning disc microscope. Upon distal junction appearance, distances were measured on the Fiji software, drawing a line between proximal and distal junction, and measuring the length of the lines.

Immunostaining

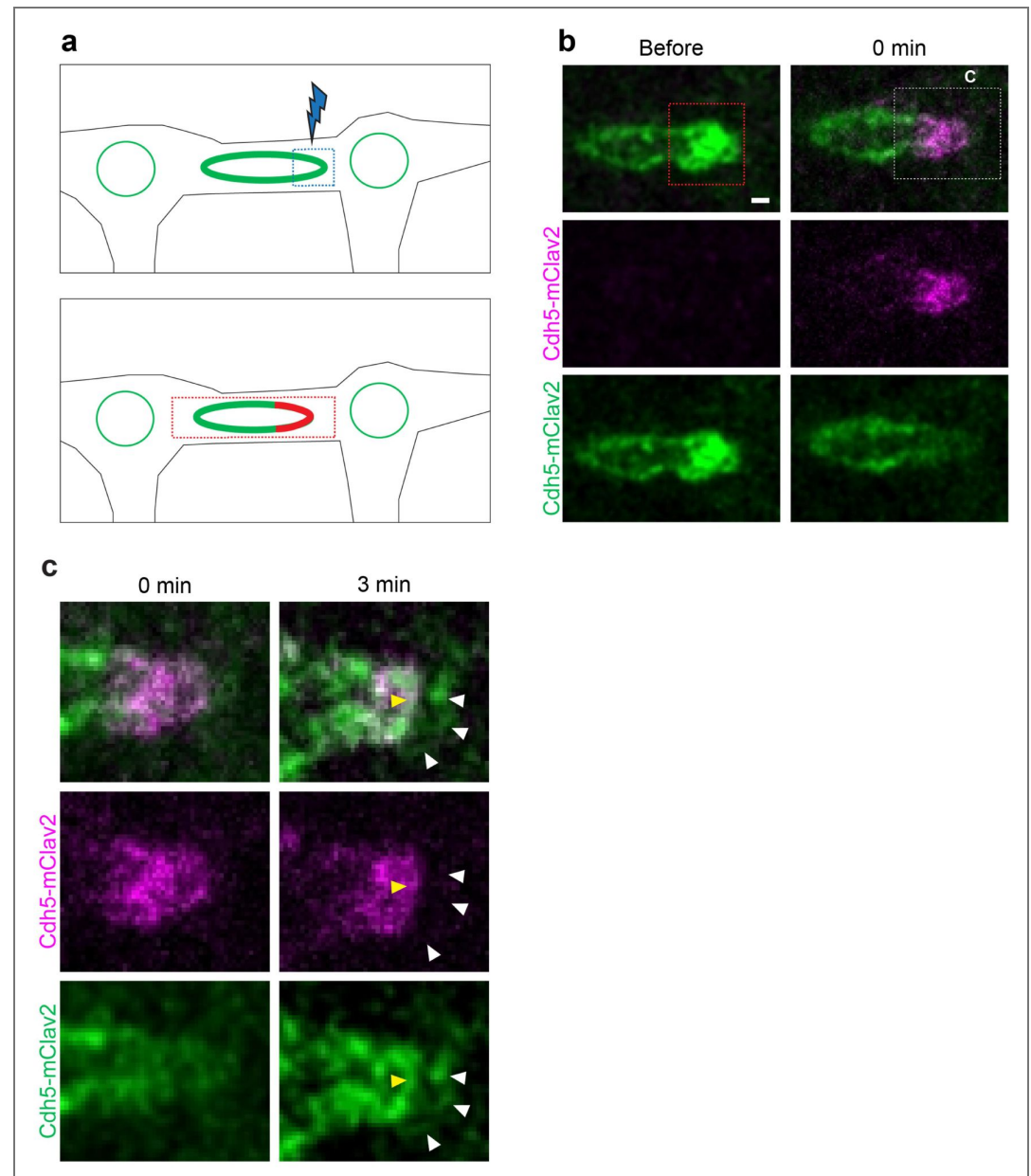
Zebrafish embryos of 30 to 32 hpf were fixed after dechoriation using Glyofixx (Thermo-Fisher Scientific) and incubated at 4°C overnight (ON) in 1.5 ml Eppendorf tubes (max. 20 embryos per tube). After fixation, embryos were washed four times for 5 minutes in 1× PBST at room temperature (RT) on a rocking platform. Embryos were then permeabilized using 0.5–1% Triton-X100 in PBST for 30 min at RT. They were then blocked with 2% BSA and 5% goat serum in PBST overnight with continuous shaking at 4°C. After blocking, the primary antibody solution was added. The primary antibodies were either diluted in blocking solution or Pierce Immunostain Enhancer Solution (Thermo Fisher Scientific). Primary antibody incubation was performed at 4°C for 3 days. After incubation, the solution was removed, and the embryos were washed at least four times for 30 minutes each with PBST at 4°C, or overnight. Subsequently, the embryos were incubated with secondary antibodies, diluted 1:2000 in blocking solution, and incubated at 4°C.

Image preparation and analysis

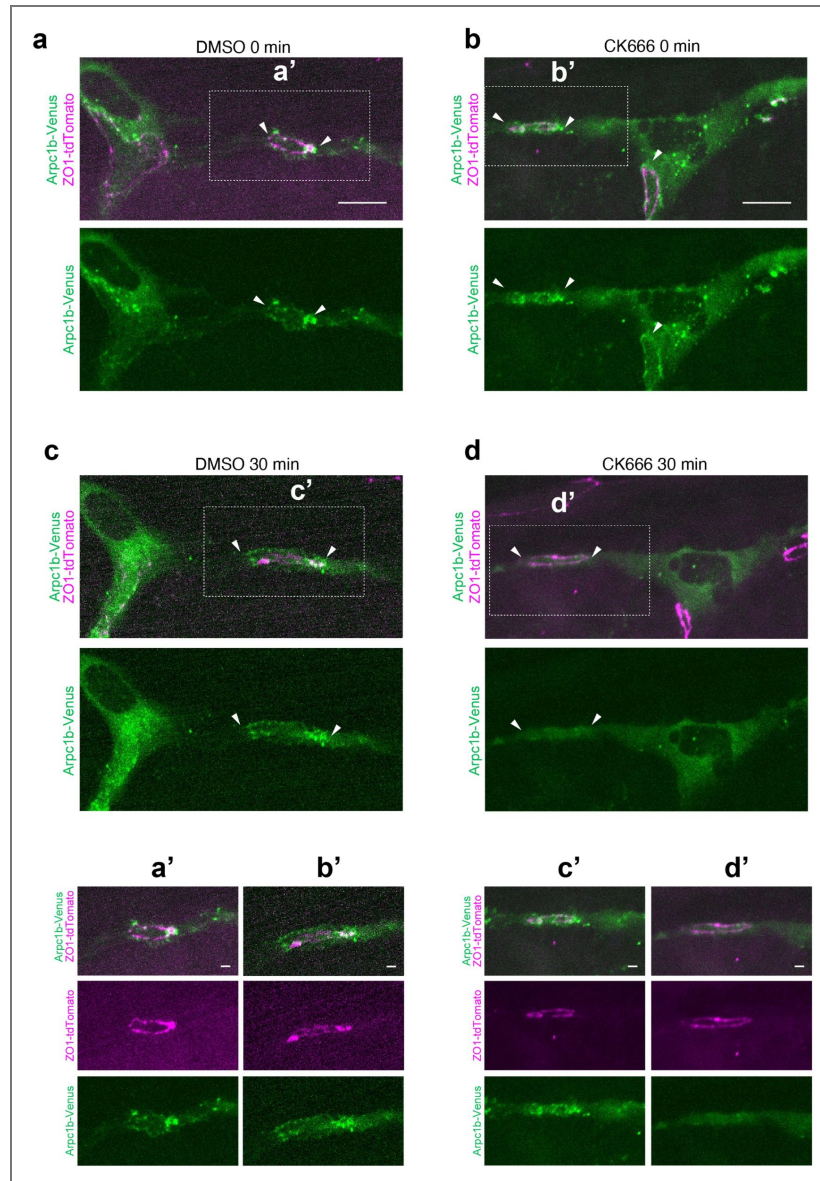
Image analysis and measurements were performed using ImageJ 2.9.0. Z-stacks were flattened by *maximum slice intensity projections*. Where needed, noise was reduced using Gaussian filtering (radius 0.7) and/or background subtracted (rolling ball radius 50) using ImageJ. Contrast and brightness of images were linearly adjusted. Image panels were created using the Open Microscopy Environment (OMERO). ARP2/3-Venus and mRuby2-UCHD localization were evaluated on a 4pixel-wide rectangle, and the average over length, corrected for background (evaluated as

average of average intensity in 5 rectangles on the cell cytoplasm) and displayed in a graph using graphpad prism. Proximal, distal, and interjunctional space domains were defined using VE-cad distribution, My9-mcherry distribution was evaluated as average intensity in the distinct domains. The intensities are corrected for background (which has been evaluated as average of average intensities in 5 rectangles in the cell cytoplasm) and displayed in a graph using graphpad prism.

Supplementary figures

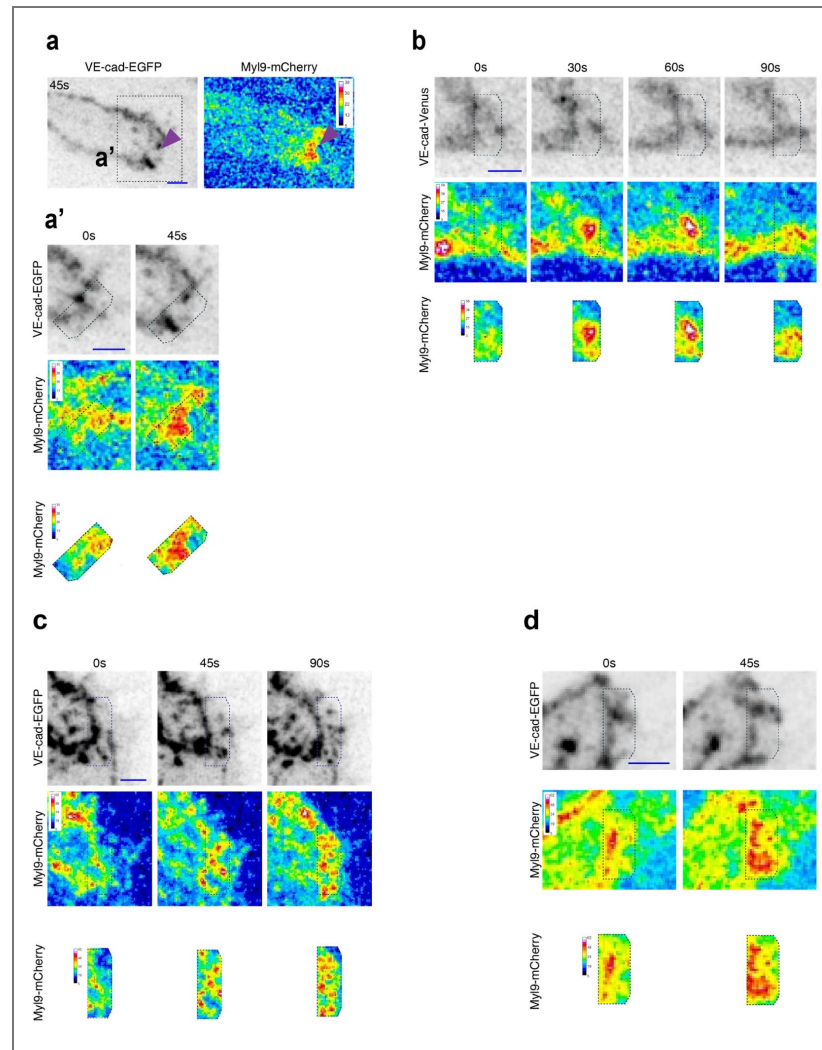


Supplementary Figure 1. The distal junctions form *de novo*. **a:** Schematic representation of the experimental design. At first mClav2 labeled VE-cad is photoconverted in one pole of the junctional ring. Then time lapse imaging is performed on the half-converted ring. **b:** Time-lapse (Supplementary movie 7) of a DLAV junctional ring of an embryo expressing Cdh5-mClav2 before and right-after photoconversion. The red dashed square delimitates the photoconverted area. White dashed square demarcates the zoomed-in area of c. **c:** Time-lapse of the zoomed in junctional ring pole, right-after photoconversion and 4 min later. White and yellow arrowheads point distal and proximal junction respectively. Distal junction is labeled by green, non-photoconverted VE-cad; but not by red photoconverted VE-cad.



Supplementary Figure 2. CK666 disrupts Arp2/3 localization in DLAV anastomotic junctions.

Stills of Arp2/3-Venus and ZO1-tdTomato before (**a**, **c**) and after 30 min treatment with DMSO 1% **c** (n=17) and CK666 200 μ M **d** (n=25). T0 around 30 hpf n>20. White arrow heads are pointing at junctional poles. **a'**, **b'**, **c'**, **d'** are magnifications of the white dashed square delimited areas.



Supplementary Figure 3. Myosin light-chain recruitment at the interjunctional space during merging.

a, a': Time-lapse of Cdh5ΔC-EGFP and Myl9b-mCherry during double junction state (Supplementary movie 17 [\[link\]](#)). **a:** Still-image showing polarized accumulation of MLC during “double junction” state. **a':** Still-images showing recruitment of MLC between proximal and distal junction. **b:** Still-images from a time-lapse (Supplementary movie 18 [\[link\]](#)) of VE-cad-Venus and Myl9b-mCherry showing increase of MLC in the junctional space prior to proximodistal junction mergence. **c, d:** Time-lapses of Cdh5ΔC-EGFP and Myl9b-mCherry (Supplementary movies 19 [\[link\]](#), 20 [\[link\]](#)) showing recruitment of MLC between proximal and distal junction during “double junction” state.

Acknowledgements

We thank Kumuthini Kulendra and Andre Rodriguez for fish care and the Imaging Core Facility of the Biozentrum (Universität Basel) for microscopy support. We thank Anne Schmid (Institute Pasteur, Paris, France) for providing *Tg(kdrl: Myl9a-GFP)ip5Tg* and Michel Bagnat (Duke University, USA) for providing *TgKI(tjp1a-tdTomato)pd1224*. This work has been supported by the Kantons Basel-Stadt and Basel-Land and by grants from the Swiss National Science Foundation (310030_200701 and 310030B_176400) to M.A..

Additional information

Data availability

The data that support the findings of this study are available in the supporting information of this article.

Statistical analysis

GraphPad Prism 10.4.1 was used to perform statistical analyses. For every experiment and analysis, sample size is specified in the figure legend. Sample sizes were not pre-determined using statistical methods, and all data meeting the quality control criteria were included in the analyses. Randomization was not applied to the experiments. Unless specified otherwise, default settings were used for all software tools in the analyses. Whenever a p-value is presented in the text or figures, the corresponding statistical test is indicated.

Author contributions

L.M., J.Y., C.W., I.P., M.A. and H.-G.B. designed the experiments. L.M., J.Y., C.W. and I.P. performed experiments and analyzed the data. J.M. and C.H. generated *Tg(fli1a:Arpc1b-mVenus-p2A-Arpc3-mTurq2)^{mr25}*. L.M. and H.-G.B. wrote the manuscript. All authors reviewed the manuscript.

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Additional files

Video 1. [JBL zo1-gfp_uchd-mRuby2](#)

Video 2. [JBL drive local lumen fusion](#)

Video 3. [JBL adjacent to apical compartment](#)

Video 4. [Formation of the distal junction \(VE-cad-Venus\)](#)

Video 5. [Distal Junction Formation \(mRuby-UCHD\)](#)

Video 6. [Distal Junction Formation \(mRuby-UCHD\)](#)

Video 7. [The distal junctions forms de novo](#)

Video 8. [Arp2/3 oscillation at junctional pole](#)

Video 9. [Arp2/3 localization at the front end of the JBL](#)

Video 10. [Arp2/3 is required for JBL formation](#)

Video 11. [Arp2/3 is required for JBL formation](#)

Video 12. [Arp2/3 is required for junction elongation](#)

- Video 13.** [🔗](#) Arp2/3 is required for junction elongation
- Video 14.** [🔗](#) Distal and proximal junctions merge
- Video 15.** [🔗](#) MLC is enriched at the junctional poles
- Video 16.** [🔗](#) Myosin light-chain dynamics correlate with junctional merging 1
- Video 17.** [🔗](#) Myosin light-chain dynamics correlate with junctional merging 2
- Video 18.** [🔗](#) Myosin light-chain dynamics correlate with junctional merging 3
- Video 19.** [🔗](#) Myosin light-chain dynamics correlate with junctional merging 4
- Video 20.** [🔗](#) Myosin light-chain dynamics correlate with junctional merging 5
- Video 21.** [🔗](#) Actomyosin contractility drives junctional conversion (DMSO)
- Video 22.** [🔗](#) Actomyosin contractility drives junctional conversion (Y-27632)
- Video 23.** [🔗](#) Actomyosin contractility drives junctional conversion (DMSO)
- Video 24.** [🔗](#) Actomyosin contractility drives junctional conversion (Y-27632)
- Video 25.** [🔗](#) Molecular Mechanism of JBL function during endothelial junction elongation

References

1. **Aleström P**, D'Angelo L, Midtlyng PJ, Schorderet DF, Schulte-Merker S, Sohm F, Warner S (2020) Zebrafish: Housing and husbandry recommendations. *Lab Anim* **54**:213-224
2. **Alves NG**, Motawe ZY, Yuan SY, Breslin JW (2018) Endothelial Protrusions in Junctional Integrity and Barrier Function. *Curr Top Membr* **82**:93
3. **Andrew DJ**, Ewald AJ (2010) Morphogenesis of epithelial tubes: Insights into tube formation, elongation, and elaboration. *Dev Biol* **341**:34-55
4. **Bagnat M**, Cheung ID, Mostov KE, Stainier DYC (2007) Genetic control of single lumen formation in the zebrafish gut. *Nat Cell Biol* **9**:954-960
5. **Betz C**, Lenard A, Belting H-G, Affolter M (2016) Cell behaviors and dynamics during angiogenesis. *Dev (Cambridge)* **143**
6. **Blaser H**, Reichman-Fried M, Castanon I, Dumstrei K, Marlow FLL, Kawakami K, Solnica-Krezel L, Heisenberg CP, Raz E (2006) Migration of Zebrafish Primordial Germ Cells: A Role for Myosin Contraction and Cytoplasmic Flow. *Dev Cell* **11**:613-627
7. **Blum Y**, Belting HG, Ellertsdóttir E, Herwig L, Lüders F, Affolter M (2008) Complex cell rearrangements during intersegmental vessel sprouting and vessel fusion in the zebrafish embryo. *Dev Biol* **316**:312-322
8. **Cai D**, Chen SC, Prasad M, He L, Wang X, Choesmel-Cadamuro V, Sawyer JK, Danuser G, Montell DJ (2014) Mechanical feedback through E-cadherin promotes direction sensing during collective cell migration. *Cell* **157**:1146-1159
9. **Campàs O**, Noordstra I, Yap AS (2024) Adherens junctions as molecular regulators of emergent tissue mechanics. *Nat Rev Mol Cell Biol* **25**:252-269
10. **Cao J**, Ehling M, März S, Seebach J, Tarbashevich K, Sixta T, Pitulescu ME, Werner AC, Flach B, Montanez E, et al. (2017) Polarized actin and VE-cadherin dynamics regulate junctional remodelling and cell migration during sprouting angiogenesis. *Nat Commun* **8**:1-20
11. **Dong B**, Horie T, Denker E, Kusakabe T, Tsuda M, Smith WC, Jiang D (2009) Tube formation by complex cellular processes in *Ciona intestinalis* notochord. *Dev Biol* **330**:237-249
12. **Eberlein J**, Herdt L, Helker CSM, Malchow J, Rittershaus A, Baumeister S (2021) Molecular and cellular mechanisms of vascular development in zebrafish. *Life* **11**
13. **Ellertsdóttir E**, Lenard A, Blum Y, Krudewig A, Herwig L, Affolter M, Belting HG (2010) Vascular morphogenesis in the zebrafish embryo. *Dev Biol* **341**:56-65

14. **Franco CA**, Jones ML, Bernabeu MO, Vion AC, Barbacena P, Fan J, Mathivet T, Fonseca CG, Ragab A, Yamaguchi TP, *et al.* (2016) Non-canonical wnt signalling modulates the endothelial shear stress flow sensor in vascular remodelling. *eLife* **5**:1-22
15. **Friedl P**, Mayor R (2017) Tuning collective cell migration by cell-cell junction regulation. *Cold Spring Harb Perspect Biol* **9**:1-18
16. **Gebala V**, Collins R, Geudens I, Phng LK, Gerhardt H (2016) Blood flow drives lumen formation by inverse membrane blebbing during angiogenesis in vivo. *Nat Cell Biol* **18**:443-450
17. **Gladysheva J**, Evnukova E, Kondakova E, Kulakova M, Efremov V (2021) Neurulation in the posterior region of zebrafish, *Danio rerio* embryos. *J Morphol* **282**:1437-1454
18. **Henson JH**, Yeterian M, Weeks RM, Medrano AE, Brown BL, Geist HL, Pais MD, Oldenbourg R, Shuster CB (2015) Arp2/3 complex inhibition radically alters lamellipodial actin architecture, suspended cell shape, and the cell spreading process.
19. **Herwig L**, Blum Y, Krudewig A, Ellertsdottir E, Lenard A, Belting HG, Affolter M (2011) Distinct cellular mechanisms of blood vessel fusion in the zebrafish embryo. *Curr Biol* **21**:1942-1948
20. **Hetrick B**, Han MS, Helgeson LA, Nolen BJ (2013) Small molecules CK-666 and CK-869 inhibit actin-related protein 2/3 complex by blocking an activating conformational change. *Chem Biol* **20**:701-712
21. **Jakobsson L**, Franco CA, Bentley K, Collins RT, Ponsioen B, Aspalter IM, Rosewell I, Busse M, Thurston G, Medvinsky A, *et al.* (2010) Endothelial cells dynamically compete for the tip cell position during angiogenic sprouting. *Nat Cell Biol* **12**:943-953
22. **Jin SW**, Beis D, Mitchell T, Chen JN, Stainier DYC (2005) Cellular and molecular analyses of vascular tube and lumen formation in zebrafish. *Development* **132**:5199-5209
23. **Legendijk AK**, Gomez GA, Baek S, Hesselson D, Hughes WE, Paterson S, Conway DE, Belting HG, Affolter M, Smith KA, *et al.* (2017) Live imaging molecular changes in junctional tension upon VE-cadherin in zebrafish. *Nat Commun* **8**
24. **Lancino M**, Majello S, Herbert S, De Chaumont F, Tinevez JY, Olivo-Marin JC, Herbomel P, Schmidt A (2018) Anisotropic organization of circumferential actomyosin characterizes hematopoietic stem cells emergence in the zebrafish. *eLife* **7**:1-36
25. **Lecuit T**, Lenne PF, Munro E (2011) Force generation, transmission, and integration during cell and tissue morphogenesis. *Annu Rev Cell Dev Biol* **27**:157-184
26. **Lenard A**, Ellertsdottir E, Herwig L, Krudewig A, Sauter L, Belting HG, Affolter M (2013) In vivo analysis reveals a highly stereotypic morphogenetic pathway of vascular anastomosis. *Dev Cell* **25**:492-506
27. **Lowery LA**, Sive H (2005) Initial formation of zebrafish brain ventricles occurs independently of circulation and requires the *nagie oko* and *snakehead/atplala.1* gene products. *Development* **132**:2057-2067
28. **Maître JL**, Heisenberg CP (2013) Three functions of cadherins in cell adhesion. *Curr Biol* **23**:626-633
29. **Malchow J**, Eberlein J, Li W, Hogan BM, Okuda KS, Helker CSM (2024) Neural progenitor-derived Apelin controls tip cell behavior and vascular patterning. *Sci Adv* **10**:1174
30. **Martinelli R**, Kamei M, Sage PT, Massol R, Varghese L, Sciuto T, Toporsian M, Dvorak AM, Kirchhausen T, Springer TA, *et al.* (2013) Release of cellular tension signals self-restorative ventral lamellipodia to heal barrier micro-wounds. *J Cell Biol* **201**:449-465
31. **Mayor R**, Etienne-Manneville S (2016) The front and rear of collective cell migration. *Nat Rev Mol Cell Biol* **17**:97-109
32. **Munson C**, Huiskens J, Bit-Avragim N, Kuo T, Dong PD, Ober EA, Verkade H, Abdelilah-Seyfried S, Stainier DYC (2008) Regulation of neurocoel morphogenesis by *Pard6yb*. *Dev Biol* **324**:41-54
33. **Navis A**, Bagnat M (2015) Developing pressures: FLUID forces driving morphogenesis. *Curr Opin Genet Dev* **32**:24-30

34. **Noordstra I**, Hermoso MD, Schimmel L, Bonfim-Melo A, Currin-Ross D, Duong CN, Kalappurakkal JM, Morris RG, Vestweber D, Mayor S, *et al.* (2023) An E-cadherin-actin clutch translates the mechanical force of cortical flow for cell-cell contact to inhibit epithelial cell locomotion. *Dev Cell* **58**:1748-1763e6
35. **Paatero I**, Sauteur L, Lee M, Lagendijk AK, Heutschi D, Wiesner C, Guzmán C, Bieli D, Hogan BM, Affolter M, *et al.* (2018) Junction-based lamellipodia drive endothelial cell rearrangements in vivo via a VE-cadherin-F-actin based oscillatory cell-cell interaction. *Nat Commun* **9**
36. **Phng L-K** (2018) Endothelial Cell Dynamics during Blood Vessel Morphogenesis. *Zebrafish, Medaka, and Other Small Fishes* 17-35
37. **Phng LK**, Belting HG (2021) Endothelial cell mechanics and blood flow forces in vascular morphogenesis. *Semin Cell Dev Biol* **120**:32-43
38. **Sauteur L**, Krudewig A, Herwig L, Ehrenfeuchter N, Lenard A, Affolter M, Belting HG (2014) Cdh5/VE-cadherin promotes endothelial cell interface elongation via cortical actin polymerization during angiogenic sprouting. *Cell Rep* **9**:504-513
39. **Schoofs H**, Daubel N, Schnabellehner S, Grönloh MLB, Martínez P, Halme S, Marks A, Jeansson AM, Barcos M, Benedito S, *et al.* (2025) Dynamic cytoskeletal regulation of cell shape supports resilience of lymphatic endothelium. *Nature* **641**:465-475
40. **Schuermann A**, Helker CSM, Herzog W (2014) Angiogenesis in zebrafish. *Semin Cell Dev Biol* **31**:106-114
41. **Strilić B**, Kučera T, Eglinger J, Hughes MR, McNagny KM, Tsukita S, Dejana E, Ferrara N, Lammert E (2009) The Molecular Basis of Vascular Lumen Formation in the Developing Mouse Aorta. *Dev Cell* **17**:505-515
42. **Taha AA**, Taha M, Seebach J, Schnittler HJ (2014) ARP2/3-mediated junction-associated lamellipodia control VE-cadherin-based cell junction dynamics and maintain monolayer integrity. *Mol Biol Cell* **25**:245-256
43. **Uehata M**, Ishizaki T, Satoh H, Ono T, Kawahara T, Morishita T, Tamakawa H, Yamagami K, Inui J, Maekawa M, *et al.* (1997) Calcium sensitization of smooth muscle mediated by a Rho-associated protein kinase in hypertension. *Nature* **389**:990-994
44. **Yin J**, Schellinx N, Maggi L, Gundel K, Wiesner C, Kotini MP, Lee M, Phng L-K, Belting H-G, Affolter M (2024) Initiation of lumen formation from junctions via differential actomyosin contractility regulated by dynamic recruitment of Rasip1. *Nat Commun* **15**:9714
45. **Zeeb M**, Strilic B, Lammert E (2010) Resolving cell-cell junctions: Lumen formation in blood vessels. *Curr Opin Cell Biol* **22**:626-632

Peer reviews

Reviewer #1 (Public review):

Summary:

Lumen formation is a fundamental morphogenetic event essential for the function of all tubular organs, notably the vertebrate vascular network, where continuous and patent conduits ensure blood flow and tissue perfusion. The mechanisms by which endothelial cells organize to create and maintain luminal space have historically been categorized into two broad strategies: cell shape changes, which involve alterations in apical-basal polarity and cytoskeletal architecture, and cell rearrangements, wherein intercellular junctions and positional relationships are remodeled to form uninterrupted conduits. The study presented here focuses on the latter process, highlighting a unique morphogenetic module, junction-based lamellipodia (JBL), as the driver for endothelial rearrangements.

Strengths:

The key mechanistic insight from this work is the requirement of the Arp2/3 complex, the classical nucleator of branched actin filament networks, for JBL protrusion. This implicates Arp2/3-mediated actin polymerization in pushing force generation, enabling plasma membrane advancement at junctional sites. The dependence on Arp2/3 positions JBL within the family of lamellipodia-like structures, but the junctional origin and function distinguish them from canonical, leading-edge lamellipodia seen in cell migration.

Weaknesses:

The study primarily presents descriptive observations and includes limited quantitative analyses or genetic modifications. Molecular mechanisms are typically interrogated through the use of pharmacological inhibitors rather than genetic approaches. Furthermore, the precise semantic distinction between JAIL and JBL requires additional clarification, as current evidence suggests their biological relevance may substantially overlap.

<https://doi.org/10.7554/eLife.109264.1.sa2>

Reviewer #2 (Public review):

Summary:

In Maggi et al., the authors investigated the mechanisms that regulate the dynamics of a specialized junctional structure called junction-based lamellipodia (JBL), which they have previously identified during multicellular vascular tube formation in the zebrafish. They identified the Arp2/3 complex to dynamically localize at expanding JBLs and showed that the chemical inhibition of Arp2/3 activity slowed junctional elongation. The authors therefore concluded that actin polymerization at JBLs pushes the distal junction forward to expand the JBL. They further revealed the accumulation of Myl9a/Myl9b (marker for MLC) at the junctional pole, at interjunctional regions, suggesting that contractile activity drives the merging of proximal and distal junctions. Indeed, chemical inhibition of ROCK activity decreased junctional mergence. With these new findings, the authors added new molecular and cellular details into the previously proposed clutch mechanism by proposing that Arp2/3-dependent actin polymerization provides pushing forces while actomyosin contractility drives the merging of proximal and distal junctions, explaining the oscillatory protrusive nature of JBLs.

Strengths:

The authors provide detailed analyses of endothelial cell-cell dynamics through time-lapse imaging of junctional and cytoskeletal components at subcellular resolution. The use of zebrafish as an animal model system is invaluable in identifying novel mechanisms that explain the organizing principles of how blood vessels are formed. The data is well presented, and the manuscript is easy to read.

Weaknesses:

While the data generally support the conclusions reached, some aspects can be strengthened. For the untrained eye, it is unclear where the proximal and distal junctions are in some images, and so it is difficult to follow their dynamics (especially in experiments where Cdh5 is used as the junctional marker). Images would benefit from clear annotation of the two junctions. All perturbation experiments were done using chemical inhibitors; this can be further supported by genetic perturbations.

<https://doi.org/10.7554/eLife.109264.1.sa1>

Reviewer #3 (Public review):

The paper by Maggi et al. builds on earlier work by the team (Paatero et al., 2018) on oriented junction-based lamellipodia (JBL). They validate the role of JBLs in guiding endothelial cell rearrangements and utilise high-resolution time-lapse imaging of novel transgenic strains to visualise the formation of distal junctions and their subsequent fusion with proximal junctions. Through functional analyses of Arp2/3 and actomyosin contractility, the study identifies JBLs as localized mechanical hubs, where protrusive forces drive distal junction formation, and actomyosin contractility brings together the distal and proximal junctions. This forward movement provides a unique directionality which would contribute to proper lumen formation, EC orientation, and vessel stability during these early stages of vessel development.

Time-lapse live imaging of VEC, ZO-1, and actin reveals that VEC and ZO-1 are initially deposited at the distal junction, while actin primarily localizes to the region between the proximal and distal sites. Using a photoconvertible Cdh5-mClav2 transgenic line, the origin of the VEC aggregates was examined. This convincingly shows that VE-cadherin was derived from pools outside the proximal junctions. However, in addition to de novo VEC derived from within the photoconverted cell, could some VEC also be contributed by the neighbouring endothelial cell to which the JBL is connected?

As seen for JAILs in cultured ECs, the study reveals that Arp2/3 is enhanced when JBLs form by live imaging of Arpc1b-Venus in conjunction with ZO-1 and actin. Therefore Arp2/3 likely contributes to the initial formation of the distal junction in the lamellopodium.

Inhibiting Arp2/3 with CK666 prevents JBL formation, and filopodia form instead of lamellopodia. This loss of JBLs leads to impaired EC rearrangements.

Is the effect of CK666 treatment reversible? Since only a short (30 min) treatment is used, the overall effect on the embryo would be minimal, and thus washing out CK666 might lead to JBL formation and normalized rearrangements, which would further support the role of Arp2/3.

From the images in Figure 4d it appears that ZO-1 levels are increased in the ring after CK666 treatment. Has this been investigated, and could this overall stabilization of adhesion proteins further prevent elongation of the ring?

To explore how the distal and proximal junctions merge, imaging of spatiotemporal imaging of Myl9 and VEC is conducted. It indicates that Myl9 is localized at the interjunctional fusion site prior to fusion. This suggests pulling forces are at play to merge the junctions, and indeed Y 27632 treatment reduces or blocks the merging of these junctions.

For this experiment, a truncated version of VEC was used which lacks the cytoplasmic domain. Why have the authors chosen to image this line, since lacking the cytoplasmic domain could also impair the efficiency of tension on VEC at both junction sites? This is as described in the discussion (lines 328-332).

Since the time-lapse movies involve high-speed imaging of rather small structures, it is understandable that these are difficult to interpret. Adding labels to indicate certain structures or proteins at essential timepoints in the movies would help the readers understand these.

<https://doi.org/10.7554/eLife.109264.1.sa0>